

Project Workflow Outline

This workflow follows a "Back-to-Front" approach, ensuring the data integrity is solid before we build the UI.

Phase 1: Data Architecture (The Foundation)

- **SQL Schema Design:** Define tables for **Users, Roles, Menus, Orders,** and **Reviews.**
- **NoSQL Strategy:** Define the structure for the "Admin Dashboard" data (likely a document-based structure for stats and logs).
- **Relationships:** Map out One-to-Many (e.g., User to Orders) and Many-to-Many (e.g., Menus to Categories/Allergens).

Phase 2: Backend Development (Python/API)

- **Environment Setup:** Virtualenv, Flask/FastAPI (or your preferred Python framework), and DB drivers.
- **Authentication Engine:** Implement JWT (JSON Web Tokens) or Session logic, including password hashing.
- **Business Logic Layer:** * Price calculation engine (Discounts + Delivery fees).
 - Status transition logic (Validation flow for orders).
- **Mailing Service:** Setup the automated trigger for welcome emails and order confirmations.

Phase 3: Frontend Development (Angular)

- **Project Scaffolding:** Angular CLI setup, Routing, and Core/Shared modules.
- **Service Layer:** Create Angular Services to communicate with your Python API.
- **State Management:** Handling user sessions and the "Shopping Cart" logic.

- **UI/UX Implementation:**

- Public views (Catalog, Home).
- Secure views (Customer Dashboard, Admin Panel).
- **Accessibility (RGAA):** Ensuring semantic HTML and ARIA labels from the start.

Phase 4: Quality Assurance & Deployment

- **Testing:** Unit tests for Python logic and Angular components.
- **Documentation Finalization:** Creating the User Manual and Technical Doc (MCD, Sequence Diagrams).
- **Deployment:** Setting up the production environment (e.g., Heroku, PythonAnywhere, or a VPS) and the SQL production DB.